



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

Course Information

Course Code: AIC270

Credit Hours: 3(2,1)

Lab Hours/Week: 3

Course Title: **Programming for Artificial Intelligence**

Lecture Hours/Week: 2

Pre-Requisites: **CSC262- Artificial Intelligence**

Course Objectives:

- To explain the fundamental constructs of programming language for data analysis and representation;
- To discuss and apply Solve and analyze programming and data analysis problems using standard libraries and/or toolboxes of the programming language.

Course Contents:

Introduction to Programming Language (Python): IDE for the Language, Variables, Expressions, Operands and Operators, Loops, Control Structures, Debugging, Error Messages, Functions, Strings, Lists, Object-Oriented Constructs & Basic Graphics in the Language; Applied Mathematics for AI; Regression Analysis & Chain Rule; Version Control (Git and Subversion); Libraries/Toolboxes Necessary for Data Analysis; Analyzing & Plotting Numerical Data (E.G., Numpy, Matplotlib, Anaconda & Pandas for Python); and Case Studies.

Unit wise Major Topics:

Unit	Topic	No. of Teaching Hours
1.	Introduction to Programming Language (Python): Integrated Development Environments (IDEs) for Python; Variables; Expressions; Operands & Operators: Basic Syntax & Operations in Python, Data Types, Arithmetic, & Logical Operations, Regular Expression; Loops & Control Structures; Functions; File Handling; Exception Handling; and Debugging.	6
2.	Python Data Structures: Strings; Lists; Tuples; and Dictionaries; Applied Mathematics for AI; and Regression Analysis & Chain Rule.	6
3.	Object-Oriented Programming in Python.	4
4.	Libraries/Toolboxes Necessary for Data Analysis: Overview of Essential Python Libraries (e.g., Numpy, Pandas, Matplotlib); Analyzing & Plotting Numerical Data: Techniques for Data analysis & Visualization.	4
5.	Object Detection and Tracking: OpenCV, Tracking Objects, Face & Eye Detection & Tracking; and AI Applications using Deep Learning.	6
6.	AI based Games; and Version Control (Git and Subversion).	4
Total Contact Hours		30

Mapping of CLOs and GAs:

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy	GA
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			Learning Level	
CLO's for Theory				
CLO-1	1-2	Apply fundamental programming constructs to solve basic computational problems.	Applying	2
CLO-2	2-4	Develop and use classes and objects, employing object-oriented programming principles to create modular and scalable code.	Understanding	2
CLO-3	5	Clarify deep learning techniques for building AI applications.	Understanding	2,3
CLO's for Lab				
CLO-4	1-6	Implement AI based applications and optimization techniques using Python.	Applying	2-5
CLO Assessment Mechanism:				
Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4
Quizzes	Quiz 1	Quiz 2&3	Quiz 4	-
Assignments	Assignment 1	Assignment 2&3	Assignment 4	Lab Assignments
Mid Term Exam	Mid Term Exam	Mid Term Exam	-	Lab Mid Term Exam
Final Term Exam	Final Term Exam			Lab Project/ Lab Final Term Exam
Text and Reference Books:				
Textbook:				
1. Artificial Intelligence with Python: Your complete guide to building intelligent apps using Python 3.x, Artasanchez, A. & Joshi, P., Packt Publishing, 2020.				
Reference Books:				
1. Introduction to Logic Programming (Synthesis Lectures on Artificial Intelligence and Machine Learning), Genesereth, M., et al., Morgan & Claypool, 2020.				
AI for Games, Millington, I., CRC Press, 2019.				